

Combinatorics PhD Exam, August 2024

1. Let $h(n)$ be the number of words of length n over the alphabet $\{a, b, c\}$ in which an a is never immediately followed by a b . Find the ordinary generating function of the sequence $\{h(n)\}$.
2. Let $H(n)$ be the number of permutations of length n that have an even number of fixed points. Find the exponential generating function of the sequence $\{H(n)\}$.
3. Select a 132-avoiding permutation π of length n uniformly at random. Which is more likely, that n is at either end of π , or that n is somewhere between positions 2 and $n - 1$ (inclusive) in π ? Justify your answer.
4. Let T be a tree. Is it true that there is a vertex v in T that is part of all paths of maximum length in T ? Justify your answer.
5. Prove that a graph in which all of the vertices have even degrees contains no bridges. (Recall that an edge e of a graph G is a bridge if $G - e$ has more connected components than G .)
6. Prove that every convex polyhedron has two faces that have the same number of vertices.
7. Find the number of all 2-element antichains in the Boolean algebra B_n .
8. Let P be a finite poset. Prove that the number of elements in a maximum chain in P equals the number of antichains in the smallest antichain cover of P . (Recall that an antichain cover of P is a set of antichains whose union is P .)